Chapter II.3

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3.1. Show that a morphism $f : X \to Y$ is locally of finite type if and only if *every* affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ can be covered by open affine subsets $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B-algebra.

Proof. One direction is clear. Suppose conversely that $f : X \to Y$ is locally of finite type, and fix $\operatorname{Spec} A \cong U \subset X$ and $\operatorname{Spec} B \cong V \subset Y$ affine, with $f(U) \subset V$. To proceed, we must first prove the following lemma.

Lemma 1. *Fix a sheaf morphism $f : X \to Y$ which is locally of finite type,* $\operatorname{Spec} A \cong U \subset X$, $\operatorname{Spec} B \cong V \subset Y$, $f(U) \subset Y$ *as above, and denote the map of rings associated to $f|_U : U \to V$ by $\varphi : B \to A$. There exist $f_{11}, \dots, f_{rs} \in A$ and $g_1, \dots, g_r \in B$ such that*

$$\operatorname{Spec} A = D(f_{11}) \cup \cdots \cup D(f_{rs}),$$
$$\operatorname{Spec} B = D(g_1) \cup \cdots \cup D(g_r),$$
$$f(D(f_{ij})) \subset D(g_i),$$

and the induced map

$$\varphi_{g_i} : B_{g_i} \to A_{f_{ij}}$$

is of finite type.

Proof. We have an affine cover $\{V_i = \operatorname{Spec} B_i\}$ of X such that each $f^{-1}(V_i)$ can be covered by open affines $U_{ij} = \operatorname{Spec} A_{ij}$, where A_{ij} is a finitely generated B_i algebra. Fix some i and write

$$V_i \cap V = \bigcup_{D(g_\beta) \subset V_i \cap V} D(g_\beta),$$

where $D(g_\beta)$ is affine in both V_i and V. Then $f^{-1}(D(g_\beta)) = D(\varphi(g_\beta))$ and fixing some j allows us to write

$$\begin{aligned} U_{ij} \cap U &= \bigcup_{D(f_\alpha) \subset U_{ij} \cap U} D(f_\alpha) \\ &= \bigcup_{\substack{D(f_\alpha) \subset U_{ij} \cap U \\ D(g_\beta) \subset V_i \cap V}} D(f_\alpha \varphi(g_\beta)), \end{aligned}$$

where each $D(f_\alpha)$ and $D(\varphi(g_\beta))$ is distinguisheed in both U and U_{ij}. We denote $f_{\alpha\beta} := f_\alpha \varphi(g_\beta)$. Then $f|_{D(f_{\alpha\beta})} : D(f_{\alpha\beta}) \to D(g_\beta)$ induces the map $\varphi_{g_\beta} : B_{g_\beta} \to A_{f_{\alpha\beta}}$.

For any $g_\beta \in B$, $f_\alpha \in A$, and $f_{\alpha\beta} \in A$ as above, we write $g'_\beta \in B_i$, $f'_\alpha \in A_{ij}$, and $f'_{\alpha\beta} \in A_{ij}$ for the corresponding elements which determine the same distinguished subset. Then

$$(B_i)_{g'_\beta} = \mathscr{O}_{V_i}(D(g'_\beta)) = \mathscr{O}_V(D(g_\beta)) = B_{g_\beta},$$

$$(A_{ij})_{f'_\alpha} = \mathscr{O}_{U_{ij}}(D(f'_\alpha)) = \mathscr{O}_U(D(f_\alpha)) = A_{f_\alpha},$$

and

$$(A_{ij})_{f'_{\alpha\beta}} = A_{f_{\alpha\beta}} = A_{f_\alpha} \otimes_B B_{g_\beta}.$$

Then A_{ij} a finitely generated B_i-algebra and A_{f_α} a cyclic A_{ij}-algebra gives that A_{f_α} is a finitely generated B_i-algebra. That is, there exists a surjective map

$$B_i[x_1, \ldots, x_{n_\alpha}] \ \to A_{f_\alpha}.$$

Right exactness of $- \otimes_B B_{g_\beta}$ gives us a surjective map

$$B_{g_\beta}[x_1, \ldots, x_{n_\alpha}] \ \to A_{f_{\alpha\beta}},$$

whence $A_{f_{\alpha\beta}}$ is a finitely generated B_{g_β}-algebra.

Combining these covers of the $V_i \cap V$ and $U_{ij} \cap U$ gives us a cover $\{D(g_\beta)\}_\beta$ of $\operatorname{Spec} B$, and covers $\{D(f_{\alpha\beta})\}_\alpha$ of each $f^{-1}(D(g_\beta)) = D(\varphi(g_\beta))$, where $\varphi_{g_\beta} : B_{g_\beta} \to A_{f_{\alpha\beta}}$ is of finite type. Since $\operatorname{Spec} B$ and $D(\varphi(g_\beta))$ are affine, we may assume each of these covers is finite, completing our result. □

Thus, it suffices to show the following lemma.

Lemma 2. *Let $\varphi : B \to A$ be a morphism of rings such that $\varphi_i : B \to A_{f_i}$ is of finite type for each $1 \leqslant i \leqslant r$, where $(f_1, \ldots, f_r) = (1)$. Then φ is of finite type.*

Proof. Take $\{a_{ij} = a'_{ij}/f_i^{e_{ij}}\}_{j=1}^{r_i}$ to be a set of generators for A_{f_i} as a B-algebra. Without loss of generality, $r_i = r$ for all $1 \leqslant i \leqslant r$. Define

$$A' := \varphi(B)[a'_{ij} \ : \ 1 \leqslant i, j \leqslant r],$$

noting that

$$A'_{f_i} = \varphi(B)[f_i^{-1}, a'_{ij} \ : \leqslant i, j \leqslant r] = A_{f_i}.$$

Now, consider the quotient A/A'. It suffices to show that $A/A' = 0$. For any prime ideal $\mathfrak{p} \in \operatorname{Spec} A$, we can choose i such that $f_i \notin \mathfrak{p}$. Then

$$\begin{aligned}
(A/A')_\mathfrak{p} &= \frac{A_\mathfrak{p}}{A'_\mathfrak{p}} \\
&= \frac{(A_{f_i})_\mathfrak{p}}{(A'_{f_i})_\mathfrak{p}} \\
&= \left(\frac{A_{f_i}}{A_{f_i}}\right)_\mathfrak{p} \\
&= 0_\mathfrak{p} \\
&= 0
\end{aligned}$$

gives us that in fact $A/A' = 0$, since being zero is a local condition. We conclude that A is a finitely generated B-algebra, as desired. □

□

3.2. A morphism $f : X \to Y$ of schemes is *quasi-compact* if there is a cover of Y by open affines V_i such that $f^{-1}(V_i)$ is quasi-compact for each i. Show that f is quasi compact if and only if for *every* open affine subset $V \subset Y$, $f^{-1}(V)$ is quasi-compact.

Proof. One implication is clear. Conversely, suppose there exists a cover $\{V_i = \operatorname{Spec} B_i\}$ of Y by open affines such that $f^{-1}(V_i)$ is quasi-compact for each i, and take an open affine $V = \operatorname{Spec} B \subset Y$. Fix i and write

$$V \cap V_i = \bigcup_j D(g_{ij}),$$

where we range over $D(g_{ij}) \subset V \cap V_i$ which are affine in both V and V_i. Then

$$V = \bigcup_{i,j} D(g_{ij}) = \bigcup_{1 \leqslant i,j \leqslant n} D(g_{ij})$$

by quasi-compactness of V. Note that $D(g_{ij}) = (V_i)_{g_{ij}}$, in the sense of (Ex. 2.16). Thus, we proceed via two brief lemmas.

Lemma 3. *Let $f : X \to Y$ be a map of schemes. For any $g \in \Gamma(\mathscr{O}_Y, Y)$, we have*

$$f^{-1}(Y_g) \cong X_{f^\sharp(g)}.$$

Proof. Recall that

$$Y_g = \{y \in Y \ : \ f_y \notin \mathfrak{m}_y \subset \mathscr{O}_{Y,y}\}.$$

Thus, we can write

$$f^{-1}(Y_g) = \{x \in X \ : \ f_{f(x)} \notin \mathfrak{m}_{f(x)} \subset \mathscr{O}_{Y,f(x)}\}.$$

$f_x^\sharp : \mathscr{O}_{Y,f(x)} \to \mathscr{O}_{X,x}$ is a local morphism, so $f_x^{\sharp\,-1}(\mathfrak{m}_x) = \mathfrak{m}_{f(x)}$. Hence, $h_{f(x)} \in \mathfrak{m}_{f(x)}$ if and only if $f_x^\sharp(h_{f(x)}) = f^\sharp(h)_x \in \mathfrak{m}_x$. This gives us

$$f^{-1}(Y_g) = X_{f^\sharp(g)},$$

as desired. □

Lemma 4. *Let $f : X \to Y$ be a map of schemes, and $\operatorname{Spec} B \cong V \subset Y$ an affine open with quasi-compact preimage $U := f^{-1}(V)$. For any $g \in B$, $f^{-1}(D(g))$ is quasi-compact.*

Proof. Take a finite cover $\{U_i\}$ of U by open affines. The proceeding lemma and (Ex. 2.16a) give us that

$$f^{-1}(D(g)) \cap U_i = U_{f^\sharp(g)} \cap U_i$$

is affine. Then

$$f^{-1}(D(g)) = \bigcup_{i=1}^{n} f^{-1}(D(g)) \cap U_i$$

is a finite union of quasi-compact spaces, and is thus quasi-compact. □

The above lemma gives us that each $f^{-1}(D(g_{ij})) \subset f^{-1}(V_i)$ is quasi-compact, whence $f^{-1}(V)$ is quasi-compact, as desired. □

3.3. (a) Show that a morphism $f : X \to Y$ is of finite type if and only if it is locally of finite type and quasi-compact.
(b) Conclude from this that f is of finite type if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V)$ can be covered by a finite number of open affines $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B-algebra.
(c) Show also if f is of finite type, then for *every* open affine subset $U = \operatorname{Spec} A \subset f^{-1}(V)$, A is a finitely generated B-algebra.

Proof. (a) One direction is clear. Suppose f is locally of finite type and quasi-compact, and fix an open affine cover of $\{V_i \cong \operatorname{Spec} B_i\}$. By (Ex. 3.1) and (Ex. 3.2), each $f^{-1}(V_i)$ is quasi-compact and covered by open affines $U_{ij} \cong \operatorname{Spec} A_{ij}$ such that A_{ij} is a finitely generated B_i-algebra. Hence, the $f^{-1}(V_i)$ are covered by finitely many such U_{ij}, and we're done.

(b) This is a corollary of (a), (Ex. 3.1), and (Ex. 3.2).

(c) This follows from Lemma 1 and Lemma 2, which were given in the proof of (Ex. 3.1). □

3.4. Show that a morphism $f : X \to Y$ is finite if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y, $f^{-1}(V) \cong \operatorname{Spec} A$ is affine with A a finite B-module.

Proof. Fix an open affine cover $\{V_i \cong \operatorname{Spec} B_i\}$ of Y such that $f^{-1}(V_i) \cong \operatorname{Spec} A_i$ is affine, and each A_i is a finite B_i-module. □

3.6. Let X be an integral scheme. Show that the local ring $\mathscr{O}_\xi$ of the generic point ξ of X is a field. It is called the *function field* of X, and is denoted by $K(X)$. Show also if $U = \operatorname{Spec} A$ is any open affine subset of X, then $K(X)$ is isomorphic to the quotient field of A.

Proof. Fix $U = \operatorname{Spec} A \subset X$ open affine. Then $\operatorname{Cl}_U(\{\xi\}) = \{\xi\}^- \cap U = U$ implies that ξ is the minimal prime ideal of U. That is, $\xi = (0) \in \operatorname{Spec} A$. Then $K(X) = A_{(0)}$ is the quotient field of A. □

3.9. *The Topological Space of a Product.* Recall that in the category of varieties, the Zariski topology on the product of two varieties is not equal to the product topology. Now, we see that in the category of schemes, the underlying point set of a product of schemes is not even the product set.

(a) Let k be a field, and let $\mathbb{A}^1_k = \operatorname{Spec} k[x]$ be the affine line over k. Show that $\mathbb{A}^1_k \times_{\operatorname{Spec} k} \mathbb{A}^1_k \cong \mathbb{A}^2_k$, and show that the underlying point set of the product is not the product of the underlying point set of the factors (even if k is algebraically closed).

(b) Let k be a field with s and t indeterminates over k. Then $\operatorname{Spec} k(s)$, $\operatorname{Spec} k(t)$, and $\operatorname{Spec} k$ are all one point spaces. Describe the product scheme $\operatorname{Spec} k(s) \times_{\operatorname{Spec} k} \operatorname{Spec} k(t)$.

Proof. (a) We have that $k[x] \otimes_k k[x] \cong k[x, y]$ as k-algebras, so $\mathbb{A}^1_k \times_{\operatorname{Spec} k} \mathbb{A}^1_k \cong \mathbb{A}^2_k$ as schemes over k. Fix $k = \mathbb{C}$, and consider the generic points of $\mathbb{A}^2_k$. A bijection of sets would give us a bijection of generic points, and thus a bijection of irreducible polynomials. This is clearly false, so that $\mathbb{A}^2_k \neq \mathbb{A}^1_k \times \mathbb{A}^1_k$ as sets.

(b) Fix $A = k(s) \otimes_k k(t)$, so that the product scheme is given by $X = \operatorname{Spec} A$. Consider the ideal $\mathfrak{p} = (s \otimes 1 + 1 \otimes t) \subset A$. Clearly $\mathfrak{p} \neq (0)$. If $\mathfrak{p} = (1)$, then we have

$$\begin{aligned} 1 \otimes 1 &= (s \otimes 1 + 1 \otimes t) \left(\sum_{i=1}^{n} \frac{f_i(s)}{p_i(s)} \otimes \frac{g_i(t)}{q_i(t)} \right) \\ &= \sum_{i=1}^{n} \frac{s f_i(s)}{p_i(s)} \otimes \frac{g_i(t)}{q_i(t)} + \frac{f_i(s)}{p_i(s)} \otimes \frac{t g_i(t)}{q_i(t)}. \end{aligned}$$

□

3.10. *Fibres of a morphism.*

(a) If $f : X \to Y$ is a morphism and $y \in Y$ a point, show that $\mathrm{sp}(X_y)$ is homeomorphic to $f^{-1}(y)$ with the induced topology.
(b) Let $X = \operatorname{Spec} k[s,t]/(s-t^2)$, $Y = \operatorname{Spec} k[s]$, and $f : X \to Y$ the morphism defined by $s \mapsto s$. If $y \in Y$ is a point $a \in K$ with $a \neq 0$, show that the fibre X_y consists of two points, with residue field k. If $y \in Y$ corresponds to $0 \in k$, show that the fibre X_y is a nonreduced one-point scheme. If η is the generic point of Y, show that X_η is a one-point scheme, whose residue field is an extension of degree two of the residue field of η. (Assume k algebraically closed.)

Proof. (a) We begin by considering the affine case, $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, and $f^\sharp(Y) = \varphi : A \to B$. An identical method to that of Ex. 2.1 gives us the following lemma.

Lemma 5. *Let A be a ring and $\phi : A \to S^{-1}A$ the inclusion into its localization. Then $\phi^* : \operatorname{Spec}(S^{-1}A) \to \operatorname{Spec} A$ is a homeomorphism onto its image in $\operatorname{Spec} A = X$. We denote this image $S^{-1}X$.*

The next lemma is easy, but worth proving.

Lemma 6. *Let $f : A \to B$ be a ring morphism, $X := \operatorname{Spec} A$, $Y := \operatorname{Spec} B$, and S a multiplicative subset of A. Then denoting $S^{-1}B = f(S)^{-1}B \cong B_A \otimes_A S^{-1}A$ and identifying spectra as in the prior lemma,*

$$S^{-1}f^* : \operatorname{Spec}(S^{-1}B) \to \operatorname{Spec}(S^{-1}A)$$

is the restrition of f^* to $S^{-1}Y$, and $S^{-1}Y = {f^*}^{-1}(S^{-1}X)$.

Proof. We have the following commutative diagram.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle \phi_A}\downarrow & & \downarrow{\scriptstyle \phi_B} \\ S^{-1}A & \xrightarrow[S^{-1}f]{} & S^{-1}B \end{array}$$

Applying the contravariant functor Spec, we get a diagram of topological spaces.

$$\begin{array}{ccc} X & \xleftarrow{f^*} & Y \\ {\scriptstyle \phi_A^*}\uparrow & & \uparrow{\scriptstyle \phi_B^*} \\ \operatorname{Spec}(S^{-1}A) & \xleftarrow[S^{-1}f^*]{} & \operatorname{Spec}(S^{-1}B) \end{array}$$

Then

$$f^*(S^{-1}Y) = \phi_A^*(S^{-1}f^*(\operatorname{Spec}(S^{-1}B)) \subset S^{-1}X$$

allows us to restrict to the following diagram.

$$\begin{array}{ccc} S^{-1}X & \xleftarrow{f^*|_{S^{-1}Y}} & S^{-1}Y \\ {\scriptstyle \phi_A^*}\uparrow & & \uparrow{\scriptstyle \phi_B^*} \\ \operatorname{Spec}(S^{-1}A) & \xleftarrow[S^{-1}f^*]{} & \operatorname{Spec}(S^{-1}B) \end{array}$$

By the preceding lemma, ϕ_A^* and ϕ_B^* are heomeomorphisms onto their image, whence the result follows. □

One last lemma precedes the proof of the desired statement.

Lemma 7. *Let $\mathfrak{a}$ be an ideal of A and let $\mathfrak{b} = \mathfrak{a}B$ be its extension in B. Let $f : A/\mathfrak{a} \to B/\mathfrak{b}$ be the homomorphism induced by f. If $\operatorname{Spec}(A/\mathfrak{a})$ is identified with its canonical image $V(\mathfrak{a}) \subset X$, and $\operatorname{Spec}(B/\mathfrak{b})$ with $V(\mathfrak{b}) \subset Y$, show that $\overline{f}^* = f^*|_{V(\mathfrak{b})}$.*

Proof. We have the following commutative diagram.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle \pi_A}\downarrow & & \downarrow{\scriptstyle \pi_B} \\ A/\mathfrak{a} & \xrightarrow[\overline{f}]{} & B/\mathfrak{b} \end{array}$$

Once again applying Spec,

$$\begin{array}{ccc} X & \xleftarrow{f^*} & Y \\ {\scriptstyle \pi_A^*}\uparrow & & \uparrow{\scriptstyle \pi_B^*} \\ \operatorname{Spec}(A/\mathfrak{a}) & \xleftarrow[\overline{f}^*]{} & \operatorname{Spec}(B/\mathfrak{b}) \end{array}$$

and restricting as before gives us our last diagram.

$$\begin{array}{ccc} V(\mathfrak{a}) & \xleftarrow{f^*|_{V(\mathfrak{b})}} & V(\mathfrak{b}) \\ {\scriptstyle \pi_A^*}\uparrow & & \uparrow{\scriptstyle \pi_B^*} \\ \operatorname{Spec}(A/\mathfrak{a}) & \xleftarrow[\overline{f}^*]{} & \operatorname{Spec}(B/\mathfrak{b}) \end{array}$$

By (Ex. 2.18), π_A^* and π_B^* are homeomorphisms onto their images $V(\mathfrak{a})$ and $V(\mathfrak{b})$, so that the result follows. □

Finally, we set $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, and fix $y = \mathfrak{p} \in \operatorname{Spec} A$. Clearly, we have homeomorphisms

$$\begin{aligned} X_{\mathfrak{p}} &= \operatorname{Spec}(B \otimes_A k(\mathfrak{p})) \\ &\cong \operatorname{Spec}(B \otimes_A A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}) \\ &\cong \operatorname{Spec}(B_{A_{\mathfrak{p}}}/\mathfrak{p}B_{A_{\mathfrak{p}}}) \\ &\cong V(\mathfrak{p}B_{A_{\mathfrak{p}}}), \end{aligned}$$

where the last equality follows from (Ex. 2.18) as before. Then equalities

$$\begin{aligned} &\cong V(\varphi_{\mathfrak{p}}(\mathfrak{m}_{\mathfrak{p}})B_{A_{\mathfrak{p}}}) \\ &\cong (\varphi_{\mathfrak{p}}^*)^{-1}(V(\mathfrak{m}_{\mathfrak{p}})) \\ &\cong f_{\mathfrak{p}}^{-1}(\mathfrak{m}_{\mathfrak{p}}) \\ &\cong f^{-1}(\mathfrak{p}). \end{aligned}$$

follow from minimality of the extension of an ideal, maximality of $\mathfrak{m}_{\mathfrak{p}} \trianglelefteqslant A_{\mathfrak{p}}$, and the second lemma respectively.

Fix affine open covers $\{U_i = \operatorname{Spec} A_i\}$ of X and $\{V_j = \operatorname{Spec} B_j\}$ of Y. Let $g : \operatorname{Spec} k(y) \to Y$ be the natural map, giving $\operatorname{Spec} k(y)$ the structure of a Y-scheme. We have

$$k(y) = \mathscr{O}_{Y,y}/m_y\mathscr{O}_{Y,y} \cong (B_j)_y/y(B_j)_y := k_j(y)$$

for all $y \in Y_j$.

Then X_y is the gluing of the spaces $f^{-1}(Y_j) \times_{Y_j} \operatorname{Spec} k_j(y)$, over all $Y_j \ni y$. We can write

$$X_i \cap f^{-1}(Y_j) = \bigcup_{\substack{g_{ijk} \in X_i \\ D_i(g_{ijk}) \subset f^{-1}(Y_j)}} D_i(g_{ijk})$$

as a union of affines. Defining $f_{ijk} := f|_{D_i(g_{ijk})} : D_i(g_{ijk}) \to Y_j$, the affine case gives us that

$$\begin{aligned} X_y &= \bigcup_{\substack{i,\, j,\, g_{ijk} \in X_i \\ D_i(g_{ijk}) \subset f^{-1}(Y_j)}} D_i(g_{ijk}) \times_{Y_j} \operatorname{Spec} k_j(y) \\ &= \bigcup_{i,\, j,\, k} D_i(g_{ijk})_y \\ &= \bigcup_{i,\, j,\, k} f_{ijk}^{-1}(y) \\ &= f^{-1}(y) \cap \bigcup_{i,\, j,\, k} D_i(g_{ijk}) \\ &= f^{-1}(y). \end{aligned}$$

(b) We have that $X_a = \operatorname{Spec} A_a$, where

$$\begin{aligned} A_a &= \frac{k[s,t]}{(s-t^2)} \otimes_{k[s]} \frac{k[s]_{(s-a)}}{(s-a)k[s]_{(s-a)}} \\ &\cong k[t] \otimes_{k[t^2]} \frac{k[t^2]}{(a-t^2)} \\ &\cong \frac{k[t]}{(a-t^2)}. \end{aligned}$$

Then $\operatorname{Spec} A_a \cong V(a-t^2) = \{(\sqrt{a}+t, \sqrt{a}-t)\} \subset \operatorname{Spec} k[t]$. If $a \neq 0$, then this is a two point set, with residue fields given by

$$\begin{aligned} k_a(\sqrt{a}-t) &= \frac{k[t]}{(\sqrt{a}-t)} \cong k, \\ k_a(\sqrt{a}+t) &= \frac{k[t]}{(\sqrt{a}+t)} \cong k. \end{aligned}$$

If $a = 0$, then $\operatorname{Spec} A_0 = \{t\}$ is singleton, and $t \in A_0$ is a nilpotent.

Finally, we consider

$$\begin{aligned} A_\eta &\cong \frac{k[s,t]}{(s-t^2)} \otimes_{k[s]} \frac{k[s]_{(0)}}{(0)k[s]_{(0)}} \\ &\cong k[t] \otimes_{k[t^2]} k(t^2) \\ &\cong k(s)[\sqrt{s}] \\ &\cong k(\sqrt{s}). \end{aligned}$$

A_η is a field, so $\operatorname{Spec} A_\eta$ is singleton. Furthermore, it is a degree two extension over $k(\eta) = k(s)$, as desired. □

3.11. *Closed Subschemes.*

(a) Closed immersions are stable under base extension: if $f : Y \to X$ is a closed immersion, and if $X' \to X$ is any morphism, then $f' : Y \times_X X' \to X'$ is also a closed immersion.

(b) If Y is a closed subscheme of an affine scheme $X = \operatorname{Spec} A$, then Y is also affine, and in fact Y is the closed subscheme determined by a suitable ideal $\mathfrak{a} \subset A$ as the image of the closed immersion $\operatorname{Spec} A/\mathfrak{a} \to \operatorname{Spec} A$. [*Hints*: First show that Y can be covered by a finite number of open affine subsets of the form $D(f_i) \cap Y$, with $f_i \in A$. By adding some more f_i with $D(f_i) \cap Y$ if necessary, show that we may assume the $D(f_i)$ cover X. Next, show that $f_1, \cdots, f_r$ generate the unit ideal of A. Then use Ex. 2.17b to show that Y is affinem and Ex. 2.18d to show that Y comes from an ideal $\mathfrak{a} \subset A$] *Note*: We will prove this with sheaves of ideals later.

(c) Let Y be a closed subset of a scheme X, and give Y the reduced induced subscheme structure. If Y' is any other closed subscheme of X with the same underlying topological space, show that the closed immersion $Y \to X$ factors through Y'. We express this property by saying that the reduced induced structure is the smallest subscheme structure on a closed subset.

(d) Let $f : Z \to X$ be a morphism. Then there is a unique closed subscheme Y of X with the following property: the morphism f factors through Y, and if Y' is any other closed subscheme of X through with f factors, then $Y \to X$ factors through Y' also. We call Y the *scheme-theoretic image* of f. If Z is a reduced scheme, then Y is just the reduced induced structure on the closure of the image $f(Z)$.

Proof. (a) Take an affine open cover $\{X_i \cong \operatorname{Spec} A_i\}$ of X and let $Y_i \cong \operatorname{Spec} B_i$, $X_i' \cong \operatorname{Spec} C_i$ be the inverse images of the X_i. This is wrong I think! $f|_{Y_i} : Y_i \to X_i$ corresponds to a map ϕ_i : Then $Y \times_X X'$ is the gluing of the open affine sets

$$U_i := Y_i \times_{X_i} X_i' \cong \operatorname{Spec}(B_i \otimes_{A_i} C_i),$$

and $f'|_{U_i} : \operatorname{Spec}(B_i \otimes_{A_i} C_i) \to \operatorname{Spec} C_i$ corresponds to the map ϕ □

3.12. *Closed subschemes of* $\operatorname{Proj} S$.

(a) Let $\varphi : S \to T$ be a surjective homomorphism of graded rings, preserving degrees. Show that the open set U is equal to $\operatorname{Proj} T$, and the morphism $f : \operatorname{Proj} T \to \operatorname{Proj} S$ is a closed immersion.

(b) If $I \subset S$ is a homogeneous ideal, take $T = S/I$ and let Y be the closed subscheme of $X = \operatorname{Proj} S$ defined as image of the closed immersion $\operatorname{Proj} S/I \to X$. Show that different homogeneous ideals can give rise to the same closed subscheme. For example, let d_0 be an integer and let $I' = \oplus_{d \geqslant 0} I_d$. Show that I and I' determine the same subscheme.
We will see later that every closed subscheme of X comes from a homogeneous ideal I of S (at least in the case where S is a polynomial ring over S_0).

Proof. (a) Denote $Y = \operatorname{Proj} S$ and $X = \operatorname{Proj} T$. We have $\varphi(S_+) = T_+$, so that $U = X$ by definition. Fix homogeneous $g \in S_+$. Surjectivity of $\varphi : S \to T$ implies $\varphi_g : S_g \to T_{\varphi(g)}$ is surjective. Then φ degree preserving gives us that $(S_g)_d = \varphi_g^{-1}((T_g)_d)$ for all $d \geqslant 0$, whence $\varphi_g : S_{(g)} \to T_{(\varphi(g))}$ is surjective. It follows that $f^\sharp : \mathscr{O}_X \to f_* \mathscr{O}_Y$ is surjective.

Surjectivity of φ gives us $\varphi(\varphi^{-1}(\mathfrak{b}))$ for all $\mathfrak{b} \trianglelefteq T$, whence

$$\mathfrak{p} \supset \mathfrak{b} \iff \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\mathfrak{b}).$$

Then

$$f(V(\mathfrak{b})) = V(\varphi^{-1}(\mathfrak{b})) \cap f(X)$$

shows that, in particular,

$$f(X) = V(\ker\varphi) \cap f(X) \subset V(\ker\varphi).$$

Take $\mathfrak{p} \in V(\ker\varphi)$, and $a \in \varphi^{-1}(\varphi(\mathfrak{p}))$. Then we have $a' \in \mathfrak{p}$ with $\varphi(a) = \varphi(a')$, whence $a - a' \in \ker\varphi \subset \mathfrak{p}$ implies $a \in \mathfrak{p}$. Thus, $\mathfrak{p} = \varphi^{-1}(\varphi(\mathfrak{p})) \in f(X)$ gives us $f(X) = V(\ker\varphi)$. We conclude that

$$f(V(\mathfrak{b})) = V(\ker\varphi + \varphi^{-1}(\mathfrak{b}))$$

is closed in $\operatorname{Proj} S$, so that f is a homeomorphism onto the closed set $f(X) = V(\ker\varphi)$.

(b) Let $\pi : S \to S/I$ and $\pi' : S \to S/I'$ be the natural projections. Since $I' \subset I = \ker\pi$, we can factor $\pi = \varphi\pi'$, where $\varphi : S/I' \to S/I$ is the identity $\varphi_d : S_d/I'_d = S_d/I_d \to S_d/I_d$ for $d \geqslant d_0$. Then Ex. 2.14(b) gives us that $\varphi^* : \operatorname{Proj}(S/I) \to \operatorname{Proj}(S/I')$ is an isomorphism of sheaves, so $(\pi')^*(\operatorname{Proj}(S/I')) = \varphi^*\pi^*(\operatorname{Proj}(S/I)) \cong \pi^*(\operatorname{Proj}(S/I))$ give the same closed subscheme structure. □

3.13. *Properties of Morphisms of Finite Type*

(a) A closed immersion is a morphism of finite type.
(b) A quasi-compact open immersion is of finite type.
(c) A composition of two morphisms of finite type is of finite type.
(d) Morphisms of finite type are stable under base extension.
(e) If X and Y are schemes of finite type over S, then $X \times_S Y$ is of finite type over S.
(f) If $X \xrightarrow{f} Y \xrightarrow{g} Z$ are two morphisms, and if f is quasi-compact, and $g \circ f$ is of finite type, then f is of finite type.
(g) If $f : X \to Y$ is a morphism of finite type, and if Y is noetherian, then X is noetherian.

Proof. (a) Let $f : X \to Y$ be a closed immersion, $\{U_i = \operatorname{Spec} A_i\}_{i\in I}$ an affine open cover of X, and fix $V = \operatorname{Spec} B \subset Y$ open affine. Then

$$\begin{aligned} f^{-1}(V) &= \bigcup_{i\in I} f^{-1}(V) \cap U_i \\ &= \bigcup_{\substack{i\in I,\ g_{ij}\in A_i \\ D_i(g_{ij})\subset f^{-1}(V)}} D_i(g_{ij}) \end{aligned}$$

is a union of open affines $D_i(g_{ij}) = \operatorname{Spec}(A_i)_{g_{ij}}$.

We have that $f^\sharp : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is surjective. That is, $f^\sharp_{\mathfrak{q}} : B_{\mathfrak{q}} \to A_{B_{\mathfrak{q}}}$ is surjective for each prime ideal $\mathfrak{p}$ of B. Then

$$\begin{aligned} f(D_i(g_{ij})) &= \{f^\sharp(V)^{-1}(\mathfrak{p}) \in \operatorname{Spec} B : \mathfrak{p} \trianglelefteqslant A_i,\ \mathfrak{p} \not\ni g_{ij}\} \\ &= \{\} \end{aligned}$$

(d) Let $f : X \to Y$ be a scheme of finite type, and $Y \to Y'$ a map of schemes. □

3.14. If X is a scheme of finite type over a field, show that the closed points of X are dense. Give an example to show that this is note true for arbitrary schemes.

3.15. Let X be a scheme of finite type over a field k (not necessarily algebraically closed).

(a) Show that the following three conditions are equivalent (in which case we say that X is *geometrically irreducible*).
   (i) $X \times_k \overline{k}$ is irreducible, where $\overline{k}$ is the algebraic closure of k.
   (ii) $X \times_k k_s$ is irreducible, where k_s denotes the sperable closure of k.
   (iii) $X \times_k K$ is irreducible for every extensions K/k.

(b) Show that the following three conditions are equivalent (in which case we say X is *geometrically reduced*).
   (i) $X \times_k \overline{k}$ is reduced.
   (ii) $X \times_k k_p$ is reduced, where k_p denotes the perfect closure of k.
   (iii) $X \times_k K$ is reduced for all extensions K/k.

(c) We say that X is *geometrically integral* if $X \times_k \overline{k}$ is integral. Note that, by (a) and (b), this implies $X = X \times_k k$ is integral. Give examples of integral schemes which are neither geometrically irreducible nor geometrically reduced.

3.16. *Noetherian Induction.* Let X be a noetherian topological space, and let $\mathscr{P}$ be a property of closed subsets of X. Assume that for any closed subset Y of X, if $\mathscr{P}$ holds for every proper closed subset of Y, then $\mathscr{P}$ holds for Y. (In particular, $\mathscr{P}$ must hold for the empty set.) Then $\mathscr{P}$ holds for X.

Proof. We induct on $\dim X$. Noetherianity allows us to write $X = \bigcup_{i=1}^{s} X_i$ for X_i irreducible. If $\dim X = 0$ We claim that, if $\mathscr{P}$ holds for each X_i, then $\mathscr{P}$ holds for X. Namely, if $Y = \bigcup_{i=1}^{s} Y_i$ with $Y_i = Y \cap Y_i$, then

We begin by showing the result for X irreducible. X irreducible implies $\dim X = n$ is finite, so we may induct on n. If $n = 0$, then the only proper closed subset of X is $\emptyset$, for which $\mathscr{P}$ holds. Hence, $\mathscr{P}$ holds for X.

Suppose now that the result holds for $n - 1$. Then, for any closed subset $Y \subsetneq X$, noetherianity allows us to write $Y = \bigcup_{i=1}^{s} Y_i$ for finitely many irreducible schemes Y_i with $\dim Y_i \leqslant \dim Y < \dim X$. By induction, $\mathscr{P}$ holds each of the Y_i, □

3.20. *Dimension.* Let X be an integral scheme of finite type over a field k (not necessarily algebraically closed). Use results about varieties to prove the following.

(a) For any closed point $p \in X$, $\dim X = \dim \mathscr{O}_p$, where for rings we always mean the Krull dimension.
(b) Let $K(X)$ be the function field of X. Then $\dim X = \text{tr.d. } K(X)/k$.
(c) If Y is a closed subset of X, then $\operatorname{codim}(Y, X) = \inf \dim \mathscr{O}_{X,p} | p \in Y$.
(d) If Y is a closed subset of X, then $\dim Y + \operatorname{codim}(Y, X) = \dim X$.
(e) If U is a nonempty open subset of X, then $\dim U = \dim X$.

Proof. □